New black holes and solitons in gauged supergravity and their scalar (quasi-)normal modes

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Introduction

It is well known that symmetry reduces the possible interaction terms for a field theory. For local supersymmetry, which is known as supergravity, the situation is dramatic because the consistency of the theory fixes $D = 11$ as the maximum number of dimensions for a supergravity theory. This theory is unique and was found by Cremmer, Julia and Scherk in 1978. The setup of this poster is the Freedman-Schwarz model [1], called $\mathcal{N} = 4$ $SU(2) \times SU(2)$ gauged supergravity. It was shown by Chamseddine and Volkov that this theory can be obtained by Kaluza-Klein dimensional reduction in $S^3 \times S^3 \times S^1$ from $D = 11$ supegravity. Here we will summarize some of our results published in [3] and [4].

The setup: $\mathcal{N} = 4$ $SU(2) \times SU(2)$ gauged supergravity

The field content of the Freedman-Schwarz model is given by one spin-1/2 spinor χ, one Rarita-Schwinger spin-3/2 field Ψ_μ. In the bosonic sector we have a metric tensor $g_{\mu\nu}$, two algebra valued gauge field A_μ, B_μ and two scalar and pseudo-scalar fields ϕ, $\mathbf{a}$. The action principle of the bosonic sector is

$$S = \int d^4x\sqrt{-g}\left[-\frac{1}{4}R - \frac{1}{4}e^{2\phi}\left(A_{i\mu\nu}A^{i\mu\nu} + B_{i\mu\nu}B^{i\mu\nu}\right) + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{g_A^2 + g_B^2}{8}e^{-2\phi} + \frac{1}{2}e^{-4\phi}\partial_\mu\mathbf{a}\partial^\mu\mathbf{a} - \frac{1}{2}\mathbf{a}\left(A_{i\mu\nu} * A^{i\mu\nu} + B_{i\mu\nu} * B^{i\mu\nu}\right)\right] .$$

The supersymmetry transformations for the spinors in a bosonic background are given by

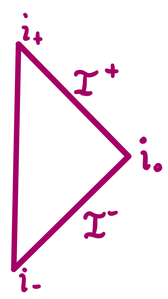
$$\delta\bar{\chi} = \bar{\epsilon}\left[\frac{i}{\sqrt{2}}\left(\not{\partial}\phi + i\gamma_5 e^{2\phi}\ \not{\partial}\mathbf{a}\right) - \frac{1}{4}e^{-\phi}C_{\mu\nu}\gamma^{\mu\nu} + \frac{1}{4}e^{\phi}\left(g_A + i\gamma_5 g_B\right)\right]$$

$$\delta\bar{\psi}_\rho = \bar{\epsilon}\left(\overleftarrow{\mathcal{D}}_\rho - \frac{i}{2}e^{2\phi}\gamma_5\partial_\rho\mathbf{a}\right) - \frac{i}{4\sqrt{2}}e^{-\phi}\bar{\epsilon}C_{\mu\nu}\gamma_\rho\gamma^{\mu\nu} + \frac{i}{4\sqrt{2}}e^{\phi}\bar{\epsilon}\left(g_A + i\gamma_5 g_B\right)\gamma_\rho$$

where $C_{\mu\nu} = A_{\mu\nu} + i\gamma_5 B_{\mu\nu}$ is an algebra valued object and $\overleftarrow{\mathcal{D}}_\rho$ is both the spacetime and the gauge convariant derivative.

1 The soliton solution

The BPS condition for planar black holes found by Klemm [2] gives in general naked singularities. Nevertheless, a double analytic continuation of these black holes gives rise new regular solutions. The metric tensor for this solution is



$$ds^2 = -\rho dt^2 + g(\rho)\,d\varphi^2 + \frac{d\rho^2}{g(\rho)} + \rho dy^2 ,$$

$$g(\rho) = \alpha\left(\rho - m - \frac{q^2}{\rho}\right) ,$$

and the matter fields are given by

$$\phi = -\ln\left(\frac{r}{2\sqrt{H_A^2 + H_B^2}}\right) , \qquad A^i_{[1]} = -\frac{H_A}{\rho}d\varphi\delta^i_3 , \qquad B^i_{[1]} = -\frac{H_B}{\rho}d\varphi\delta^i_3 .$$

The spacetime starts at ρ_0, which is $g(\rho_0) = 0$, and the coordinate φ is identified as $\varphi \sim \varphi + 4\pi/g'(\rho_0)$. It is 1/4-BPS when $m = 0$.

2 Meron ansatz for $SU(2)$

A non-Abelian self-gravitating configurations may be construct with the meron ansatz

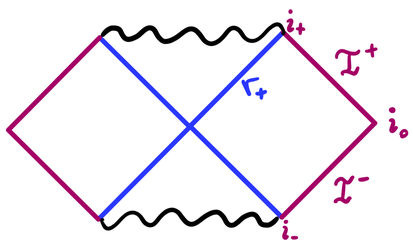
$$A_{[1]} = \lambda U^{-1}_{(\alpha)}dU_{(\alpha)} \qquad \text{where} \qquad U \in SU(2)$$

We will consider $SU(2)$ as the gauge group. A *purely magnetic configuration* can be obtained by considering the following parametrization

$$U^{\pm 1}_{(\alpha)} = \mathbf{1}\cos\zeta(r) \pm \sin\zeta(r)\,\hat{x}^i\alpha_i . \qquad \left(\text{for simplicity } \zeta \equiv \tfrac{\pi}{2}\right)$$

2.1 Double meron black hole

The double meron configuration corresponds to consider the meron ansatz in both gauge fields. The matter field and the metric tensor are given by



$$\phi(r) = -\ln\left(\frac{e_A e_B r}{\sqrt{2}\sqrt{e_A^2 + e_B^2}}\right) ,$$

$$A_{[1]} = -\frac{1}{e_A}U^{-1}_{(\alpha)}dU_{(\alpha)} , \quad B_{[1]} = -\frac{1}{e_B}U^{-1}_{(\beta)}dU_{(\beta)} ,$$

$$ds^2 = r^2\left(\tilde{\Lambda}^2 - \frac{\mu}{r^2}\right)dt^2 - \frac{dr^2}{\left(\tilde{\Lambda}^2 - \frac{\mu}{r^2}\right)} - r^2\left(d\theta^2 + \sin^2\theta d\varphi^2\right)$$

The solution acquires a conformal killing vector at spatial infinity $r\partial_r$ and the BPS condition implies $\mu = 0$, which is a naked singularity. The temperature $T = \frac{\tilde{\Lambda}^2}{2\pi}$ has the intriguing property of being independent of μ. Thus, heat capacity diverges. $\tilde{\Lambda}$ is a function of the gauge couplings.

3 Exact quasi-normal modes

It is possible to integrate in an exact manner a scalar field non-minimally coupled to the Ricci scalar with the following action principle

$$I[\Phi] = \int_{\mathcal{M}} d^4x\sqrt{-g}\left(-\frac{1}{2}\nabla_\mu\Phi\nabla^\mu\Phi - \frac{1}{2}\xi R\Phi^2\right)$$

in some manifold $\mathcal{M}$ presented here. The local isometries of the spacetimes admit variable separation. The parametrization of the time-dependent funciton is $e^{-i\omega t}$. Here I will write the spectrum of the scalar field for black holes with flat and spherical base manifold with $\gamma = 0, 1$, respectively. There are two branches for the spectrum:

$$\omega_p = -\left[(2\gamma - 8\alpha)\xi + 2\sqrt{2\xi}(1+2p)\alpha - 4p(1+p)\alpha + k^2\right]\frac{(1+2p+\sqrt{2\xi})}{(4\xi - 2(1+2p)^2)}i ,$$

$$\omega_q = -\left[(2\gamma - 8\alpha)\xi - 2\sqrt{2\xi}(1+2q)\alpha - 4q(1+q)\alpha + k^2\right]\frac{(1+2q-\sqrt{2\xi})}{4\left(1+2q+\sqrt{2\xi}\right)}i .$$

where $p, q \in \{1, 2, 3, \dots\}$. These are valid solutions when the following bound is fulfilled

$$\left[(1 - 4\xi + (2n+1)^2)\alpha + 2\gamma\xi + k^2 \mp (2n+1)\alpha\sqrt{2\xi}\right]\left(1 + 2n \mp \sqrt{2\xi}\right)^{-1} < 0$$

where the upper sign corresponds to $n \to p$ and the lower sign $n \to q$. The frequencies and the conditions are independent of the horizon radius. Hence, these spacetimes are isospectral from the point of view of the scalar field.

4 Beyond this supergravity

Extending our analysis beyond the Freedman-Schwarz supegravity in the context of the Einstein-Yang-Mills dilaton theory with the following self-interaction potential $V(\phi) = \xi e^{\sqrt{z-1}\phi} + \frac{2(z-1)}{\eta^2(z-2)}e^{\frac{\phi}{\sqrt{z-1}}} - \frac{(z+2)(z+1)}{\ell^2}$ for the dilaton and a different dilatonic coupling, we found asymptotically topologically Lifshitz black holes with interesting causal structure.

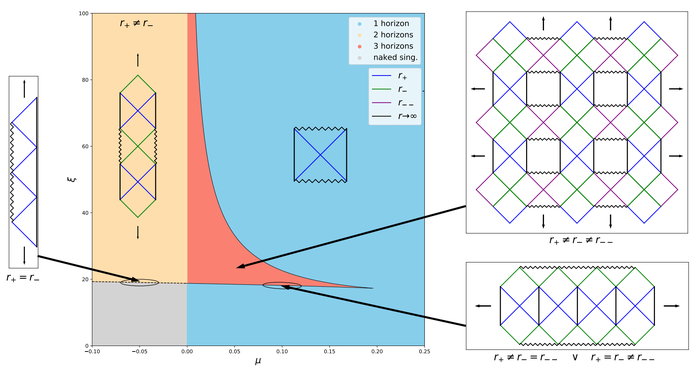


Figure 1: Causal structures of asymptotically topologically Lifshitz black holes for different values of the parameter ξ and the integration constant μ for $z = 3/2$.

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